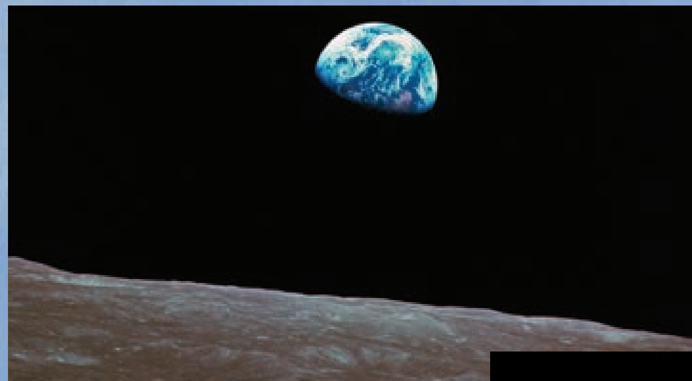
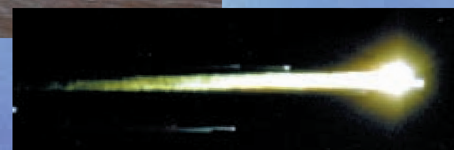
craft in Earth orbit. At the same time the powerful N-1 booster – the Soviet equivalent of Saturn V – was being readied for a test launch. More worrying for NASA, in September 1968, an unmanned Zond spacecraft, Zond 5, was sent around the Moon and brought back to Earth. Cosmonauts Alexei Leonov and Oleg Makarov began training for a circumlunar flight in the hopes that the Soviets could claim another first. With the loss of Komarov, the fact that the N-1 rocket had never tested successfully, and the return of Zond 5 at such a high trajectory, which would have killed any crew, the Soviets were more cautious. Another unmanned flight, Zond 6, was staged in November to improve the chances of bringing the cosmonauts back alive. As the Soviets hesitated, the Americans jumped in.

NASA had planned to follow Apollo 7 with another mission in Earth orbit, to test the lunar module and docking procedures. Apollo 9 would then go into lunar orbit, carrying out all the procedures for the ultimate Moon mission short of an actual Moon landing. But the desire not to be upstaged by the Soviet Union led to a change of plan. Because the lunar module for Apollo 8 was unavailable, NASA would proceed directly to a manned flight into lunar orbit.

Frank Borman, Jim Lovell, and William Anders blasted off on board Apollo 8 on December 21, 1968. They were the first crew to experience launch by the mighty Saturn V – Apollo 7 had been lifted into orbit by a Saturn 1B. Shaken and stirred by the power of the booster, within 11 minutes they had reached Earth orbit. Three hours later the Saturn's third-stage J-2 engine reignited to send them off into space. The first humans to go beyond Earth orbit watched their home planet dwindle behind them through the three-day 250,000-mile (400,000-km) voyage to the Moon.

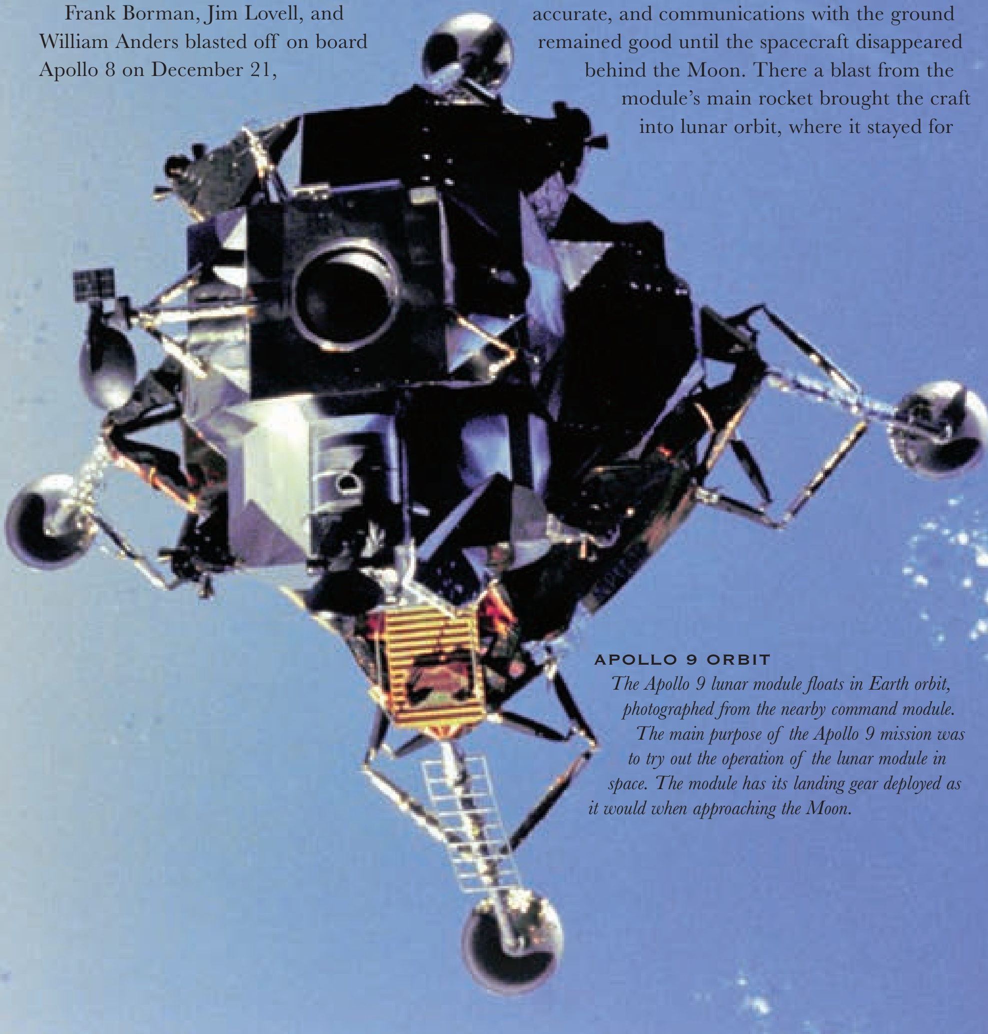
The course they took through space was accurate, and communications with the ground remained good until the spacecraft disappeared behind the Moon. There a blast from the module's main rocket brought the craft into lunar orbit, where it stayed for the following 20 hours. The event was capped by a media spectacular. Millions of viewers watched on television as black-and-white images of the astronauts and of the surface of the Moon passing below their spacecraft were broadcast live to the world. They heard the crew give their impressions of what Lovell described as "the vast loneliness" of space. As it was Christmas Eve, the astronauts all read from the Book of Genesis, and Borman wished a Happy Christmas to "all of you on the good Earth." Apollo 8 then returned safely to splash down on December 27, bringing with it what became one of the most famous photographs





HEAVENLY PHOTOS
This breathtaking photograph of the Earth seen from lunar orbit, with the surface of the Moon in the foreground, was taken by astronaut Bill Anders from the Apollo 8 module on Christmas Eve, 1968. The photograph below, showing Apollo 8 reentering the Earth's atmosphere like a meteor, was taken from a tracking aircraft.

> "The Earth from here is a grand oasis in the big vastness of space."
>
> **JIM LOVELL**
> ASTRONAUT ON APOLLO 8 SPEAKING LIVE ON TELEVISION, DECEMBER 24, 1968



APOLLO 9 ORBIT
The Apollo 9 lunar module floats in Earth orbit, photographed from the nearby command module. The main purpose of the Apollo 9 mission was to try out the operation of the lunar module in space. The module has its landing gear deployed as it would when approaching the Moon.